Classification Algorithm
K-Nearest Neighbors(KNN)

Key Concept

Definition
K-Nearest Neighbors (KNN) is a supervised machine learning algorithm used for classification and regression. In classification, KNN predicts the class of a new observation based on the classes of its **nearest neighboring observations**.

1 Introduction to KNN

K-Nearest Neighbors is one of the simplest machine learning algorithms used for classification.

Unlike Logistic Regression, KNN does not learn a mathematical equation that directly separates the classes.

Instead, KNN makes a prediction by looking at the **training observations that are closest to the new observation**.

The basic idea is:

Nearby observations are likely to belong to similar classes

For example, suppose we want to classify a new student as **Pass** or **Fail**.
If the students most similar to the new student are mostly **Pass**, KNN predicts:

Pass

2 Why is it Called K-Nearest Neighbors?

The name K-Nearest Neighbors consists of three important terms:

- **K** → Number of neighbors considered
- **Nearest** → Closest observations
- **Neighbors** → Training observations near the new observation

For example, if:

$$K = 3$$

the algorithm considers the **3 nearest training observations** to make the prediction.
If:

$$K = 5$$

the algorithm considers the **5 nearest training observations**.

Important

The value of K determines how many neighboring observations are used for making the prediction.

3 Basic Idea of KNN

Suppose we have training data containing two classes:

Class A

and

Class B

Now, consider a new observation whose class is unknown.
KNN performs the following basic process:

1. Select the value of K.
2. Calculate the distance between the new observation and all training observations.
3. Identify the K nearest observations.
4. Observe their class labels.
5. Count the number of observations belonging to each class.
6. Assign the class with the highest number of neighbors.

This is called **Majority Voting**.

4 A Simple KNN Example

Consider the following training observations:

Student	**Hours Studied**	**Result**
P_1	2	Fail
P_2	3	Fail
P_3	5	Pass
P_4	6	Pass
P_5	7	Pass

Suppose a new student has studied:

4 hours

Assume:

$$K = 3$$

The three nearest observations are:

$$3, \quad 5, \quad 2$$

Their corresponding classes are:

$$\text{Fail}, \quad \text{Pass}, \quad \text{Fail}$$

Therefore:

$$\text{Fail} = 2$$

$$\text{Pass} = 1$$

Since Fail has the majority:

$$\boxed{\text{Predicted Class} = \text{Fail}}$$

5 Distance in KNN

KNN needs to determine which training observations are closest to the new observation.

Therefore, it uses a **distance measure**.

The distance tells us how far two observations are from each other.

A smaller distance means that the observations are more similar in terms of their feature values.

A larger distance means that the observations are more different.

Key Concept

Key Idea
KNN uses distance to identify the nearest neighbors of a new observation.

6 Euclidean Distance

The most commonly used distance measure in KNN is **Euclidean Distance**.

For two observations having two features:

$$P = (x_1, y_1)$$

and

$$Q = (x_2, y_2)$$

the Euclidean distance is:

$$\boxed{d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}}$$

This represents the straight-line distance between two points.

7 Example of Euclidean Distance

Consider two observations:

$$P = (2, 3)$$

and

$$Q = (5, 7)$$

The Euclidean distance is:

$$d(P,Q) = \sqrt{(5-2)^2 + (7-3)^2}$$

$$d(P,Q) = \sqrt{3^2 + 4^2}$$

$$d(P,Q) = \sqrt{9+16}$$

$$d(P,Q) = \sqrt{25}$$

Therefore:

$$\boxed{d(P,Q) = 5}$$

Hence, the distance between the two observations is 5.

8 Distance with Multiple Features

In real-world machine learning problems, an observation usually has multiple features.
Suppose two observations are:

$$P = (p_1, p_2, \ldots, p_n)$$

and:

$$Q = (q_1, q_2, \ldots, q_n)$$

The Euclidean distance is:

$$\boxed{d(P,Q) = \sqrt{\sum_{i=1}^{n}(p_i - q_i)^2}}$$

For example, if student performance is described using:

- Hours studied
- Attendance
- Assignment score

then all these features can contribute to the distance calculation.

9 KNN Classification Example

Consider a dataset containing two features:

$$X_1 = \text{Study Hours}$$

$$X_2 = \text{Attendance}$$

Suppose the training data contains:

Student	Features	Class
P_1	$(2, 60)$	Fail
P_2	$(3, 65)$	Fail
P_3	$(5, 75)$	Pass
P_4	$(6, 80)$	Pass
P_5	$(7, 85)$	Pass

Now consider a new student:

$$P_{\text{new}} = (5, 78)$$

Suppose:

$$K = 3$$

The algorithm calculates the distance between P_{new} and every training observation.
The three observations having the smallest distances are selected.
Suppose their classes are:

$$\text{Pass}, \quad \text{Pass}, \quad \text{Fail}$$

Therefore:

$$\text{Pass} = 2$$

$$\text{Fail} = 1$$

By majority voting:

$$\boxed{\text{Predicted Class} = \text{Pass}}$$

10 Majority Voting

Majority voting is the decision-making process used by KNN classification.
Suppose:

$$K = 5$$

and the five nearest neighbors have the following classes:

Neighbor	Class
N_1	A
N_2	A
N_3	B
N_4	A
N_5	B

The class frequencies are:

$$A = 3$$

$$B = 2$$

Therefore:

$$\boxed{\text{Predicted Class} = A}$$

because Class A has the majority.

11 Choosing the Value of K

The choice of K is important because it affects the behavior of the KNN model.

For example:

$$K = 1$$

means that only the nearest training observation is considered.

On the other hand:

$$K = 10$$

means that the ten nearest observations are considered.

The value of K controls how locally or broadly the model makes its decision.

12 Small Value of K

Consider a very small value of K, such as:

$$K = 1$$

The model looks at only one nearest neighbor.

This makes the model highly sensitive to individual observations and noise.

For example, if the nearest observation happens to be incorrectly labelled, the prediction can also become incorrect.

Important

A very small value of K can make the model sensitive to noise and may lead to **overfitting**.

13 Large Value of K

Now consider a large value of K.

For example:

$$K = 20$$

The algorithm considers many neighboring observations.

This produces a smoother decision because individual observations have less influence.

However, if K is too large, the model may ignore important local patterns.

Important

A very large value of K can make the model too simple and may lead to **underfitting**.

14 Practical Selection of K

There is no single value of K that is always best.

The value is generally selected experimentally using model validation techniques.

Commonly, several values of K are tested, such as:

$$K = 1, 3, 5, 7, 9, \ldots$$

For binary classification, odd values of K are often preferred because they reduce the possibility of a tie between two classes.

For example:

$$K = 5$$

can produce:

3 votes for Class A

and:

2 votes for Class B

which gives a clear majority.

15 Importance of Feature Scaling

Feature scaling is particularly important for KNN because KNN uses distance calculations.

Consider two features:

- Age: 18–60
- Annual Income: *Rs.*2,00,000–*Rs.*20,00,000

The numerical scale of income is much larger than the scale of age.
As a result, income can dominate the distance calculation.
This may cause the model to give excessive importance to one feature.
Therefore, feature scaling is commonly applied before using KNN.
Common scaling methods include:

- Standardization
- Min-Max Scaling

Important

Since KNN is a distance-based algorithm, feature scaling can have a significant effect on its performance.

16 KNN Classification Algorithm

The basic KNN classification algorithm can be summarized as follows:

1. Choose the value of K.
2. Take the new observation whose class needs to be predicted.
3. Calculate the distance between the new observation and every training observation.
4. Sort the training observations according to their distance from the new observation.
5. Select the K nearest observations.
6. Identify the class of each selected neighbor.
7. Count the number of neighbors belonging to each class.
8. Select the class having the highest number of neighbors.
9. Assign that class to the new observation.

The complete process is:

$$\boxed{\text{New Observation} \rightarrow \text{Calculate Distances} \rightarrow \text{Find K Neighbors} \rightarrow \text{Majority Voting} \rightarrow \text{Predicted Class}}$$

17 KNN for Multiclass Classification

KNN can also be used for multiclass classification.

Suppose there are three classes:

$$A, \quad B, \quad C$$

For:

$$K = 5$$

suppose the nearest neighbors are:

$$A, \quad A, \quad B, \quad C, \quad A$$

The class frequencies are:

$$A = 3$$

$$B = 1$$

$$C = 1$$

Therefore:

$$\boxed{\text{Predicted Class} = A}$$

The same majority voting principle is used even when there are more than two classes.

18 KNN as a Lazy Learner

KNN is commonly described as a **Lazy Learning** algorithm.

This is because KNN does not build an explicit mathematical model during training in the same way as algorithms such as Logistic Regression.

Instead, it stores the training observations.

When a new observation is given, KNN performs the distance calculations and determines the nearest neighbors.

Therefore:

$$\boxed{\text{Training is simple}}$$

but:

$$\boxed{\text{Prediction can be computationally expensive}}$$

especially when the dataset contains a large number of observations.

19 Advantages of KNN

The major advantages of KNN are:

- Simple and easy to understand.
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- Easy to implement.
- Can be used for classification and regression.
- Can handle binary and multiclass classification.
- No complex model equation is required.
- Useful when similar observations tend to have similar target values. ‘‘‘

20 Limitations of KNN

KNN also has some limitations:

- Prediction can be slow for large datasets.
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- Distance calculations can become expensive when there are many observations.
- Feature scaling is important.
- Performance can be affected by irrelevant features.
- Choosing an inappropriate value of K can lead to overfitting or underfitting.
- KNN can become less effective when the number of features becomes very large. ‘‘‘

21 KNN vs Logistic Regression

Both KNN and Logistic Regression can be used for classification, but their approaches are different.

Feature	Logistic Regression	KNN
Approach	Learns a model	Uses nearest observations
Prediction	Probability-based	Distance-based
Training	Learns coefficients	Stores training data
Distance	Not required	Required
Feature Scaling	Often useful	Very important
Interpretability	Relatively high	Lower
Prediction Speed	Generally faster	Can be slower

Important

Main Difference
Logistic Regression learns a relationship between features and the target class.
KNN predicts the class by looking at the nearest training observations.

22 Important Terms

Term	Meaning
K	Number of neighbors considered
Neighbor	A training observation close to the new observation
Distance	Measure of how far two observations are
Euclidean Distance	Common distance measure used in KNN
Majority Voting	Selecting the class with the most neighbors
Feature Scaling	Bringing features to comparable numerical scales
Lazy Learner	Algorithm that performs most computation during prediction
Overfitting	Model becomes too sensitive to training observations
Underfitting	Model is too simple to capture important patterns

23 Summary

Key Concept

- KNN is a **supervised machine learning algorithm**.
  ‘‘‘
- KNN can be used for both **classification and regression**.
- In classification, KNN predicts the class of a new observation using its nearest neighbors.
- K represents the number of neighbors considered.
- Distance measures are used to identify the nearest observations.
- Euclidean Distance is one of the most commonly used distance measures.
- KNN classification uses **majority voting**.
- A very small K can lead to overfitting.
- A very large K can lead to underfitting.
- Feature scaling is important because KNN is a **distance-based algorithm**.
- KNN can handle both binary and multiclass classification. ‘‘‘

New Data Point → Distance Calculation → K Nearest Neighbors → Majority Voting → Predicted Cla